

Simulation Based Inference

NASA AI/ML STIG
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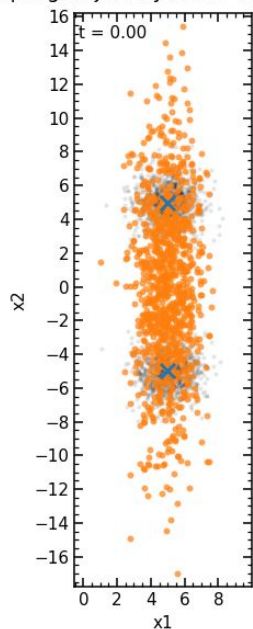
Where this talk sits in the STIG schedule?

Module 5: Generative Models:

- Mar 16 - Normalizing Flows - Gregory Green
 - Normalizing Flows, Continuous Normalizing Flows (CNF), conditional models
- Mar 23 - Diffusion Models - Duo Xu
 - Denoising Diffusion Probabilistic Models, efficient sampling, model as a learned prior
- Mar 30 - Flow Matching - Tomasz Róžański
- **Apr 6 - Simulation Based Inference (SBI) - Tomasz Róžański**
 - Based on Deistler et al., 2025 (Simulation-Based Inference: A Practical Guide, <https://arxiv.org/abs/2508.12939v1>)

One unifying picture: a probability path

Sampling trajectory of the learned flow



$$\dot{x}_t = u_t(x_t)$$

$$\partial_t p_t(x) = -\nabla \cdot (p_t(x_t) u_t(x_t))$$

Why do astronomers care about probability distributions?

$$p(x)$$

- Spectra, images, time series
- Abundances
- Initial Mass Function
- CMD
- ...

$$p(x | \theta)$$

Forward
population/simulator
models

$$p(\theta | x)$$

Parameter inference /
SBI

Bayes' theorem

$p(x)$

- Spectra, images, time series
- Abundances
- Initial Mass Function
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- ...

$p(x | \theta)$

Forward
population/simulator
models

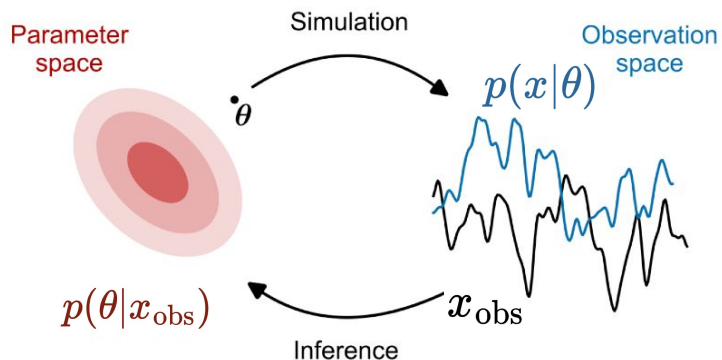
$p(\theta | x)$

Parameter inference /
SBI

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)}$$

Bayesian inference

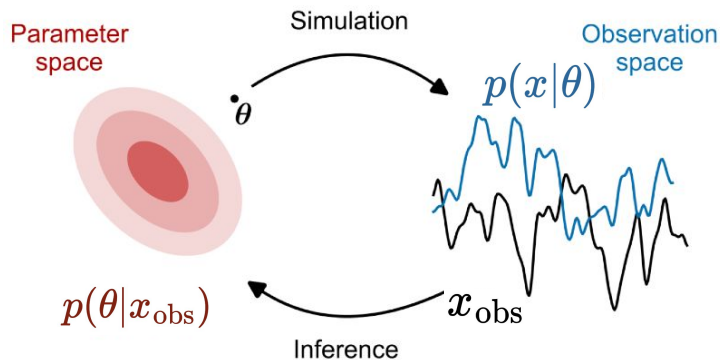
$$p(\theta|x_{\text{obs}}) = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}} = \frac{p(x_{\text{obs}}|\theta)p(\theta)}{p(x_{\text{obs}})}$$



Bayesian inference

Why SBI exists?

$$p(\theta|x_{\text{obs}}) = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}} = \frac{p(x_{\text{obs}}|\theta)p(\theta)}{p(x_{\text{obs}})}$$



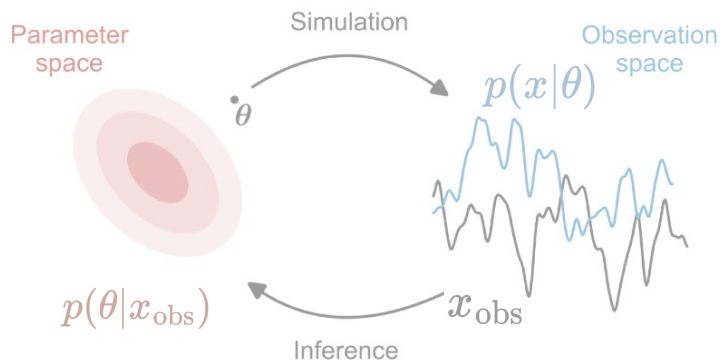
Bayesian inference

For many simulators, the likelihood is unavailable or intractable.

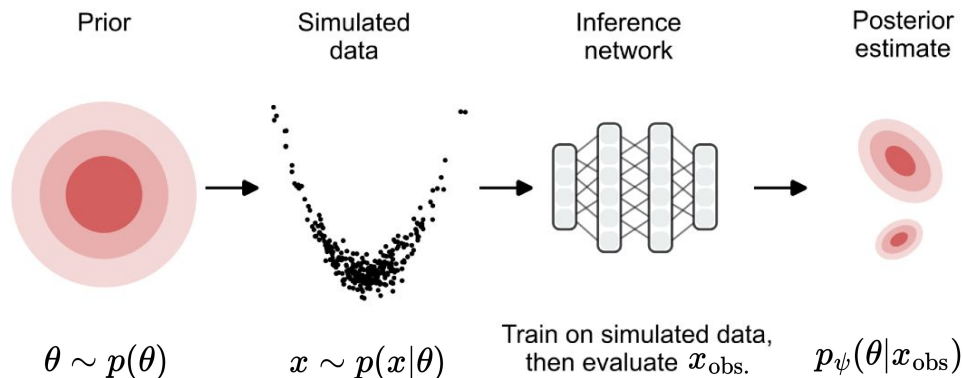
We can sample $x \sim p(x|\theta)$, but we cannot evaluate $p(x|\theta)$.

What is SBI?

$$p(\theta|x_{\text{obs}}) = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}} = \frac{p(x_{\text{obs}}|\theta)p(\theta)}{p(x_{\text{obs}})}$$

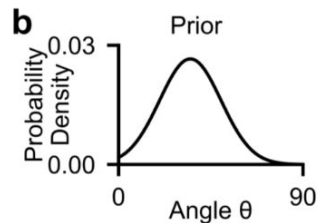
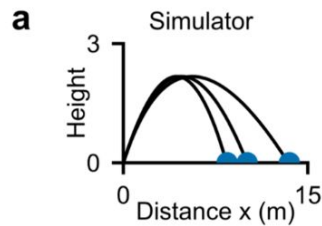


Bayesian inference



Simulation Based Inference (SBI)

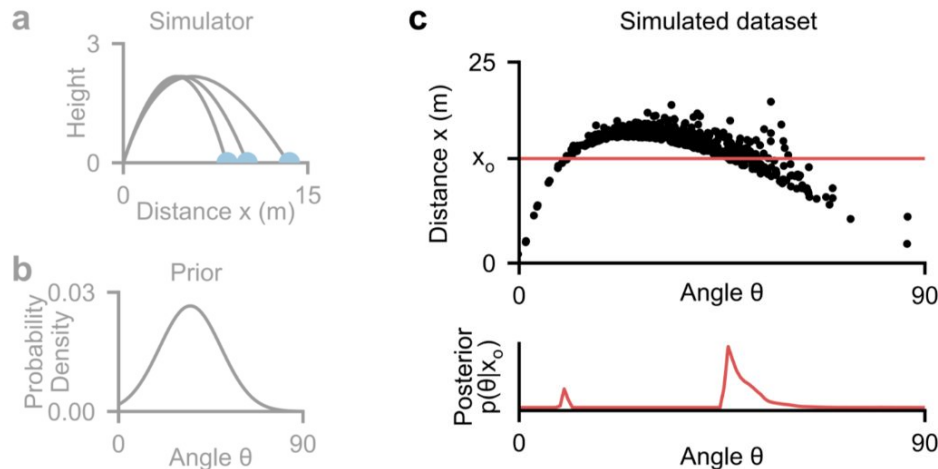
Intuition with the ball-throw example



$$\theta \sim p(\theta)$$

$$x \sim p(x|\theta)$$

Intuition with the ball-throw example

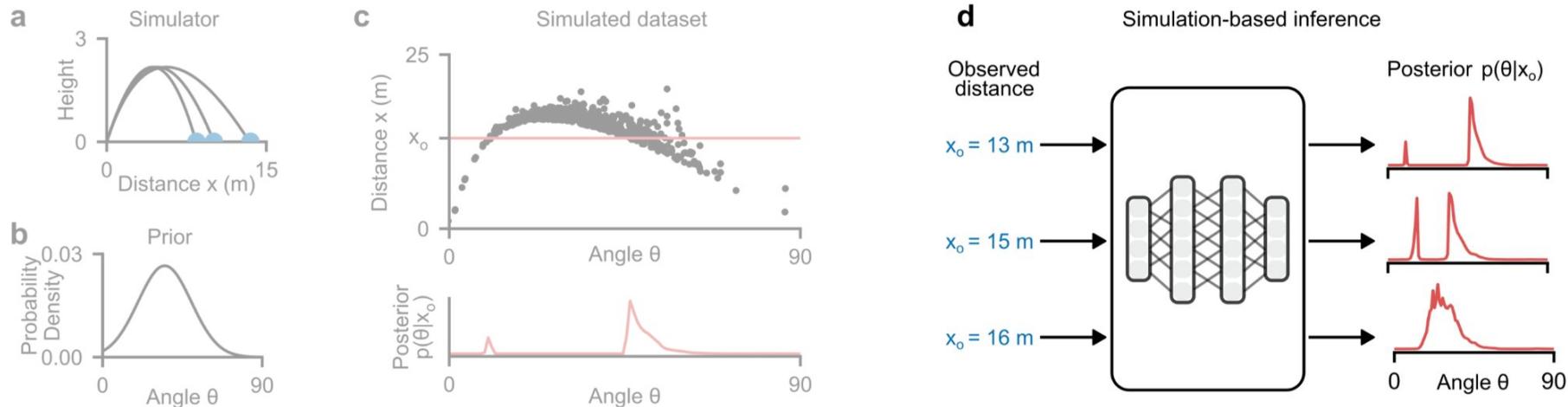


$$\theta \sim p(\theta)$$

$$x \sim p(x|\theta)$$

$$p(\theta|x_{\text{obs}})$$

Intuition with the ball-throw example



$$\theta \sim p(\theta)$$

$$x \sim p(x|\theta)$$

$$p(\theta|x_{\text{obs}})$$

$$p_{\psi}(\theta|x_{\text{obs}})$$

What does neural SBI actually learn?

Neural Posterior Estimation (NPE)

$$p(\theta|x_{\text{obs}}) = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}} = \frac{p(x_{\text{obs}}|\theta)p(\theta)}{p(x_{\text{obs}})}$$

What does neural SBI actually learn?

Neural Posterior Estimation (NPE)

Neural Likelihood Estimation (NLE)

$$p(\theta|x_{\text{obs}}) = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}} = \frac{p(x_{\text{obs}}|\theta)p(\theta)}{p(x_{\text{obs}})}$$

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Neural Ratio Estimation (NRE)

The end-to-end SBI workflow

Step 1

Define the problem:
Simulator + prior

Step 2

Choose components:
Representation, method and network

Step 3

Execute:
Run simulations, training and inference

Step 4

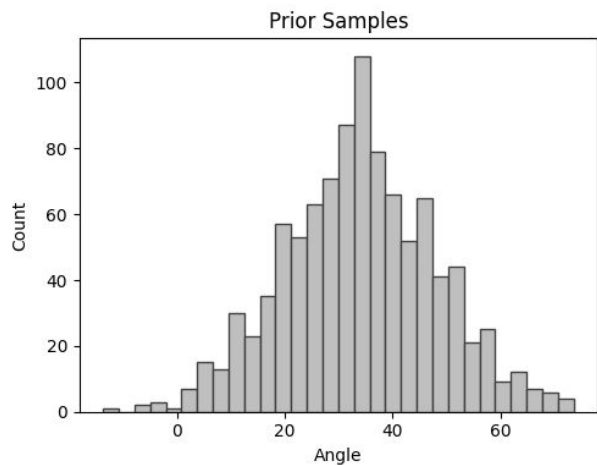
Diagnose SBI quality

Step 5

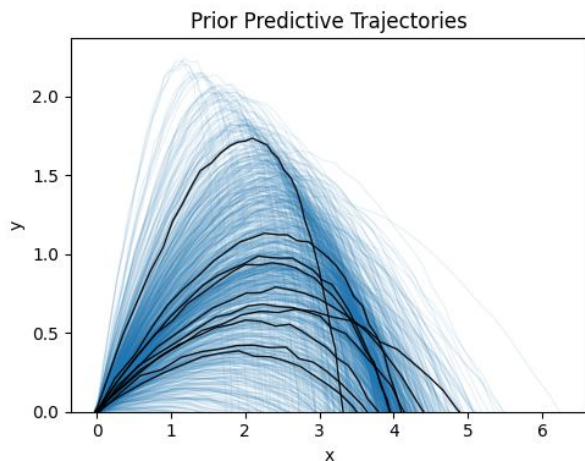
Interpret inferred posteriors



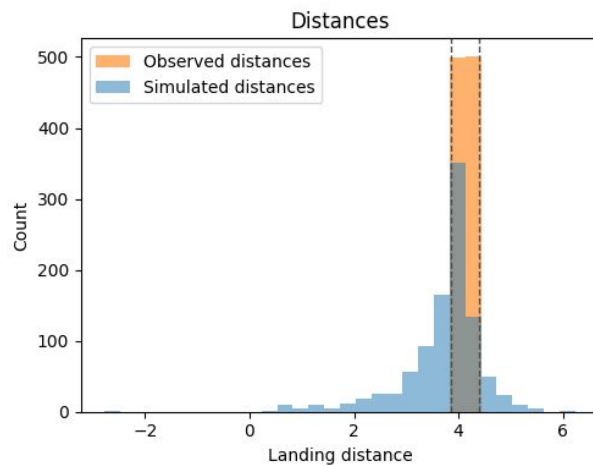
Step 1: Simulator, prior, and prior predictive checks



$$\theta \sim p(\theta)$$



$$x \sim p(x|\theta)$$



$$\{x\} \sim p(x|\theta) \quad \text{vs} \quad \{x_{\text{obs}}\}$$

Step 2: Designing the SBI pipeline

Data representation

1. Raw data
2. Summary statistics
3. Trainable embedding

SBI target

1. Posterior (NPE)
2. Likelihood (NLE)
3. Likelihood / Evidence ratio (NRE)

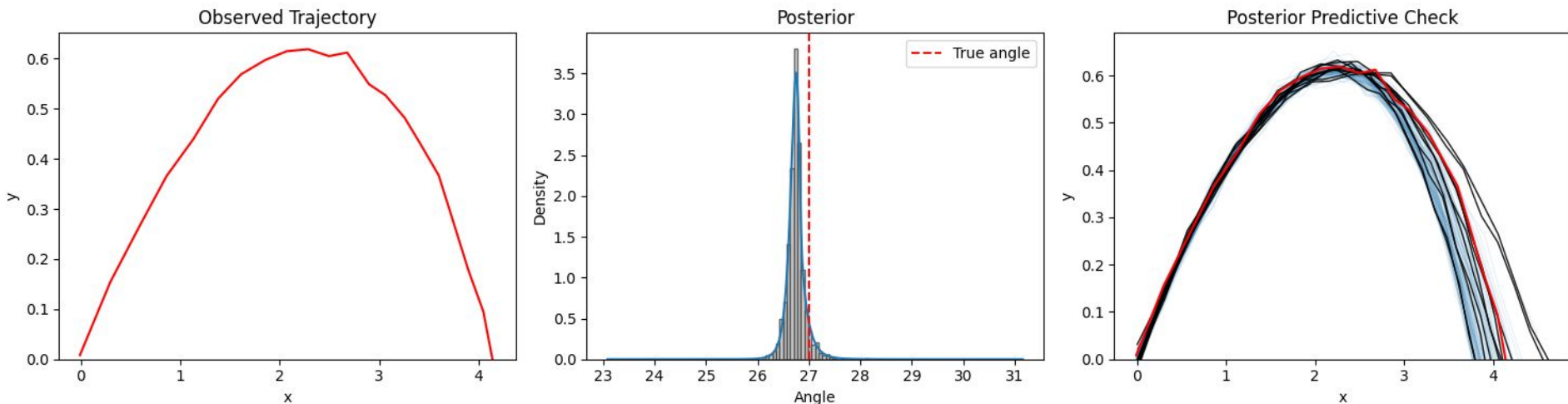
Inference network

1. Normalizing Flow (NF)
2. Continuous Normalizing Flow (CNF)
3. ResNet

Step 3: Running SBI - simulation, training, and inference

- Run simulator to create simulations bank.
- Follow good neural network (NN) training practices:
 - train/validation/test split,
 - NN architecture and training hyperparameters tuning,
 - Monitor training/validation learning curves for overfitting,
 - Well tuned NN and training often matters more than particular method.
- Start with simple minimal baseline - add complexity only if necessary.

Step 4: Posterior distribution diagnostics

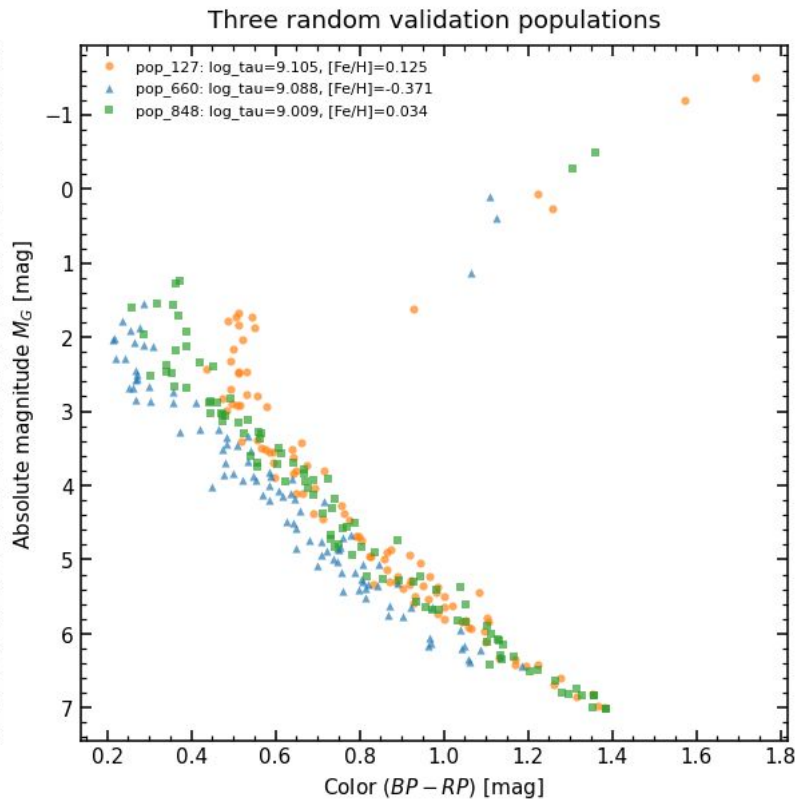
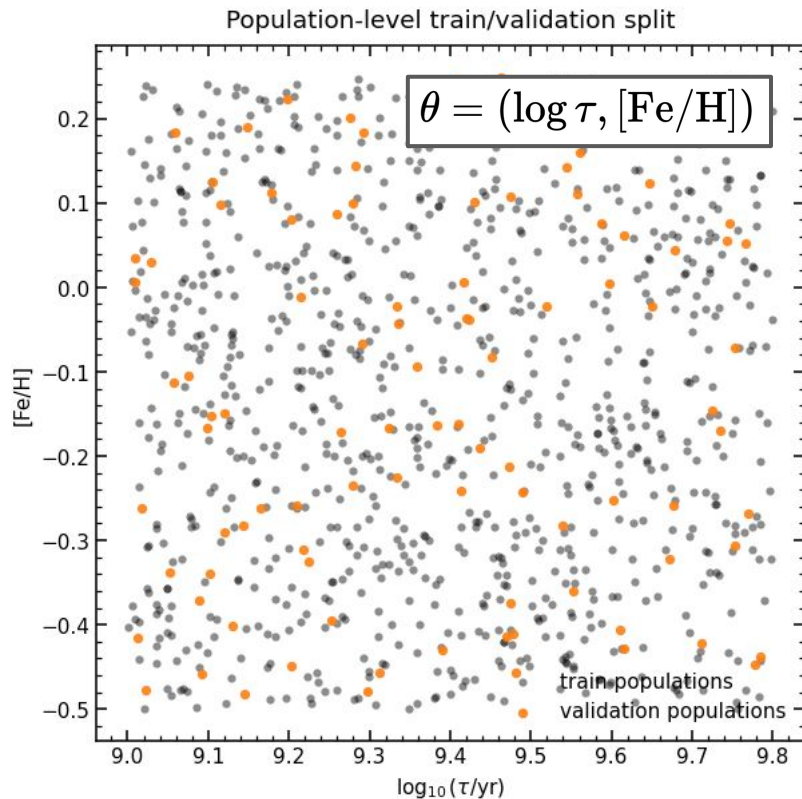


- Posterior Predictive Checks for a few samples (both simulated and real measurements)
- Global calibration tests, e.g. Simulation Based Calibration
- Local calibration tests, e.g. Local Coverage Test, Local Classifier-two-sample testing

Step 5: Interpreting the posterior

- **Visualize the posterior:** examine marginals, correlations, and corner plot.
- **Identify what the data learned:** determine which parameters are well constrained and which are still prior-driven.
- **Improve the representation:** revise summary statistics or embeddings if important information is being lost.
- **Improve the model:** use posterior structure and posterior predictive failures to guide simulator/model refinement.

Notebook example (1/4)

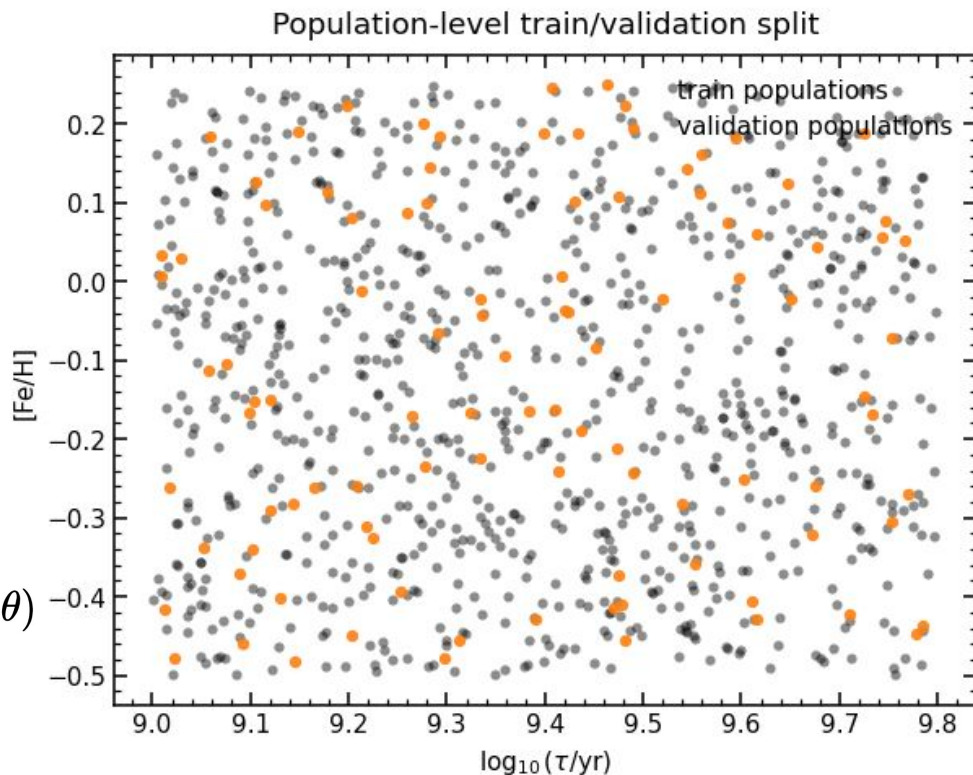


Notebook example (2/4)

$$\theta = (\log \tau, [\text{Fe}/\text{H}])$$

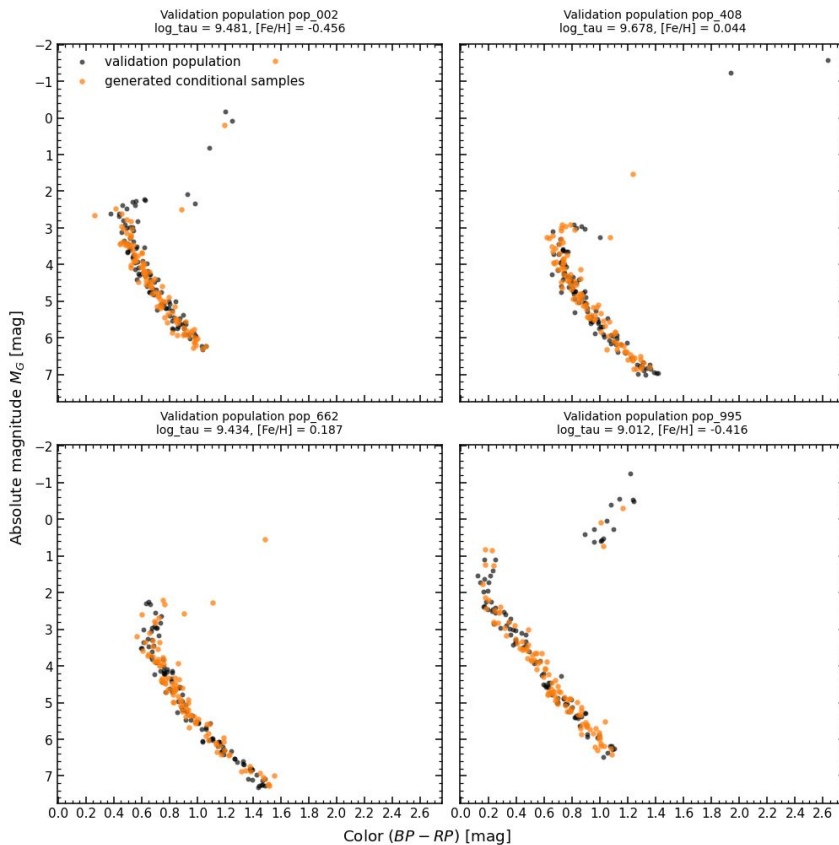
Because the samples are independent and identically distributed (i.i.d.), we use NLE parameterized by a normalizing flow.

$$p(\theta|x_{\text{obs}}) = p(\theta|x_1, x_2, \dots, x_N) = \prod_{i=1}^N p(x_i|\theta) \cdot p(\theta)$$



Notebook example (3/4)

Example samples from trained conditional normalizing flow.

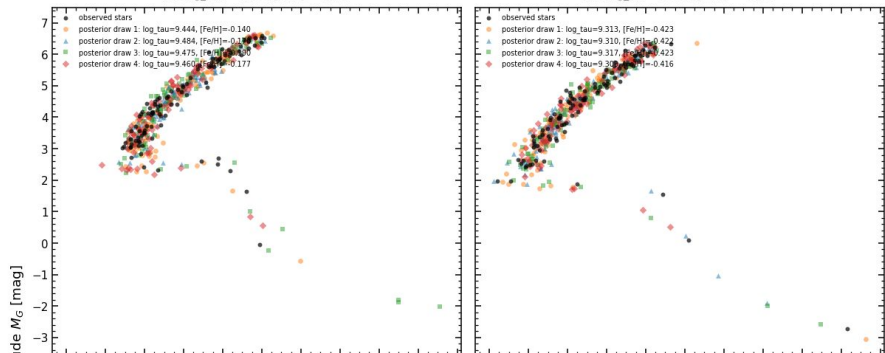


Notebook example (4/4)

Posterior predictive checks: 4 posterior draws and 100 generated stars per draw

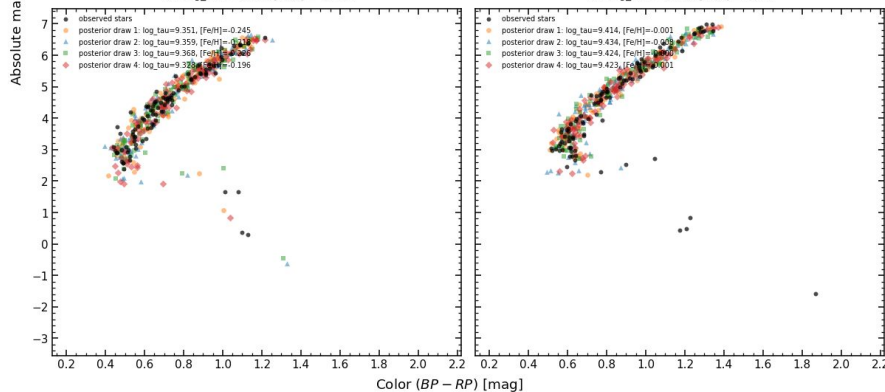
Validation example pop_025
true $\log_{10} \tau = 9.473$, $[\text{Fe}/\text{H}] = -0.213$

Validation example pop_151
true $\log_{10} \tau = 9.312$, $[\text{Fe}/\text{H}] = -0.456$

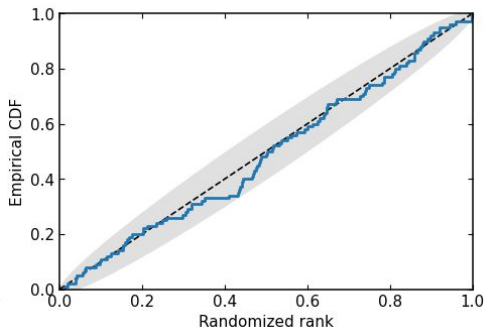
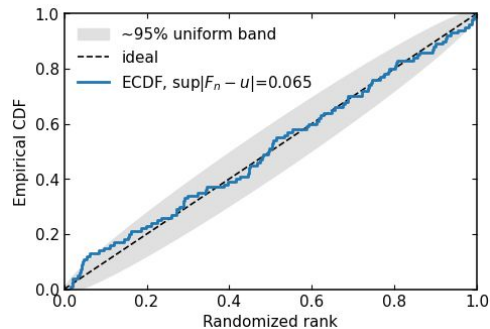
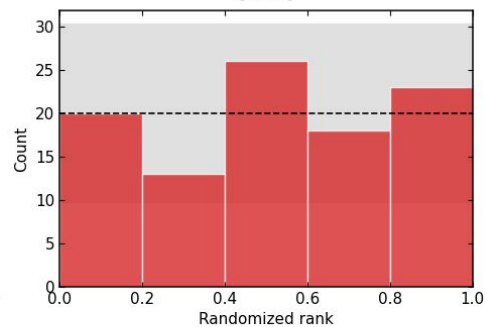
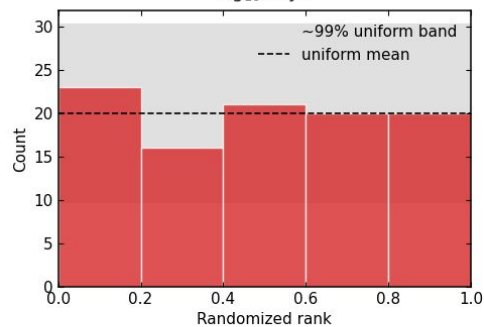


Validation example pop_650
true $\log_{10} \tau = 9.334$, $[\text{Fe}/\text{H}] = -0.225$

Validation example pop_856
true $\log_{10} \tau = 9.417$, $[\text{Fe}/\text{H}] = 0.007$



Standard marginal SBC on validation populations ($N=100$, posterior draws=180)
 $\log_{10}(\tau/\text{yr})$ $[\text{Fe}/\text{H}]$



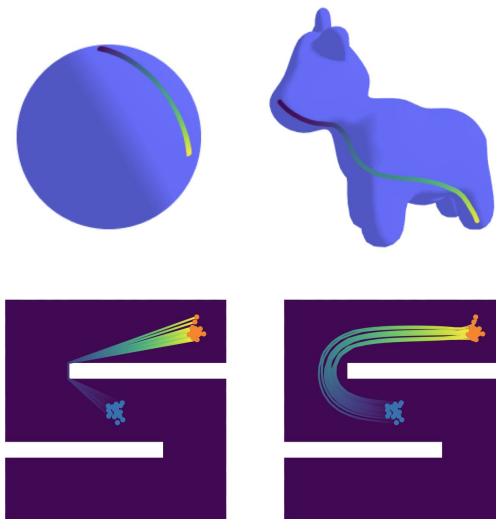
Summary and recommendations

- **SBI turns simulation into inference:** when the likelihood is unavailable or intractable, use simulations from the forward model to learn or approximate the posterior over parameters.
- **Start with NPE:** use NPE with normalizing flows as the default baseline.
- **Prefer NLE for i.i.d. data:** especially when simulations are costly, dataset size varies, and MCMC is acceptable.
- **Watch emerging alternatives:** score-based and flow-matching methods are promising when stronger i.i.d. support is needed, though inference is typically slower.
- **Cross-check important results:** whenever possible, compare at least two methods, such as NPE and NLE.

Question from the last week

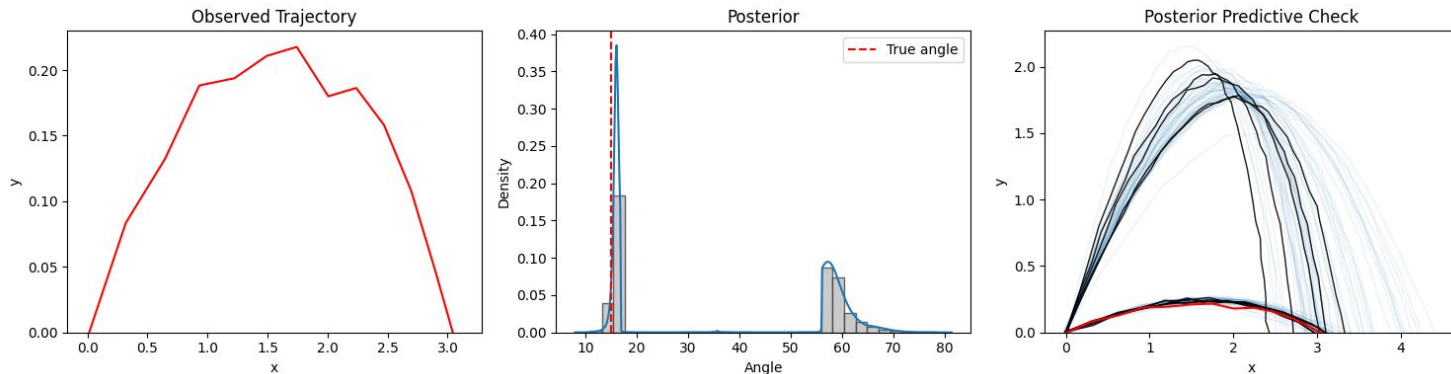
When conditional tracks different than straight lines are useful?

- Riemannian Flow Matching: Flow machining on general geometries



Extending summary statistics

Distance:



Distance
and
flight time:

